

Data: Current data D ; historical data $D_0 = \{D_{0h} : h = 1, \dots, H\}$; grid of points $\alpha = \{0 = \alpha_1 < \alpha_2 < \dots < \alpha_T = 1\}$; number of PP MCMC samples M_h ; number of posterior samples M .

Result: M posterior samples under the NPP in (??).

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1 for  $h \leftarrow 1$  to  $H$  do
2   for  $t \leftarrow 1$  to  $T$  do
3     Obtain  $M_h$  MCMC samples from the PP in (??) using historical
      data  $D_{0h}$ , initial prior  $\pi_0^{1/H}$ , and discounting parameter  $\alpha_t$ .
4     Conduct bridge sampling to obtain an estimate  $\widehat{Z_h(\alpha_t)}$  of  $Z_h(\alpha_t)$ 
      using the  $M_h$  samples from the prior.
5   end
6   Use linear interpolation on  $\{\widehat{Z_h(\alpha_1)}, \dots, \widehat{Z_h(\alpha_T)}\}$  to obtain a
      piecewise linear approximation  $\hat{Z}_h(\cdot)$  of  $Z_h(\cdot)$ .
7   Draw  $M$  posterior samples under the NPP by substituting  $\hat{Z}_h(\cdot)$  for
       $Z_h(\cdot)$  in (??).
8 end
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